

STANDARDS AND INFORMATION DOCUMENTS

AES41-1-2012
rev. AES41-2009
reaffirmed 2017



AES standard for digital audio - Audio-embedded metadata - Part 1: General

Users of this standard are encouraged to determine if they are using the latest printing incorporating all current amendments and editorial corrections. Information on the latest status, edition, and printing of a standard can be found at:
<http://www.aes.org/standards>

AUDIO ENGINEERING SOCIETY, INC.
551 Fifth Avenue, New York, New York 10176, US.



The AES Standards Committee is the organization responsible for the standards program of the Audio Engineering Society. It publishes technical standards, information documents and technical reports. Working groups and task groups with a fully international membership are engaged in writing standards covering fields that include topics of specific relevance to professional audio. Membership of any AES standards working group is open to all individuals who are materially and directly affected by the documents that may be issued under the scope of that working group.

Complete information, including working group scopes and project status is available at <http://www.aes.org/standards>. Enquiries may be addressed to standards@aes.org

The AES Standards Committee is supported in part by those listed below who, as Standards Sustainers, make significant financial contribution to its operation.



Focusrite®



THE TELOS ALLIANCE®



audio-technica



This list is current as of 2018/4/18

AES standard for digital audio - Audio-embedded metadata - Part 1: General

Published by

Audio Engineering Society, Inc.

Copyright © 2009, 2012, 2017 by the Audio Engineering Society

Abstract

AES41 provides for the carriage of audio metadata by embedding it in the audio samples themselves. This tightly associates the metadata with the audio, yet makes it fragile so that changes to the audio will invalidate the metadata. Several metadata sets have been defined, covering applications such as cascaded compression (bit-rate reduction), and loudness control.

This Part specifies the method by which a metadata set is embedded in the audio samples. Other parts define the content and format of the metadata sets.

An AES standard implies a consensus of those directly and materially affected by its scope and provisions and is intended as a guide to aid the manufacturer, the consumer, and the general public. The existence of an AES standard does not in any respect preclude anyone, whether or not he or she has approved the document, from manufacturing, marketing, purchasing, or using products, processes, or procedures not in agreement with the standard. Prior to approval, all parties were provided opportunities to comment or object to any provision. Attention is drawn to the possibility that some of the elements of this AES standard may be the subject of patent rights other than those identified herein. AES shall not be held responsible for identifying any or all such patent rights. Approval does not assume any liability to any patent owner, nor does it assume any obligation whatever to parties adopting the standards document. Recipients of this document are invited to submit, with their comments, notification of any relevant patent rights of which they are aware and to provide supporting documentation. This document is subject to periodic review and users are cautioned to obtain the latest edition.

Contents

0 Introduction	5
0.1 Rationale for this standard	5
0.2 Conventions used in this standard	5
0.3 Patents	5
1 Scope	6
2 Normative references	6
3 Definitions, symbols, and abbreviations	6
3.1 Definitions	6
3.2 Symbols	6
3.3 Abbreviations	7
4 Method for describing the bit-stream syntax	8
5 Specification of the audio coder control data bit-stream syntax	9
6 Semantics of the bit-stream syntax	10
6.1 Header	10
6.2 AES41 metadata type registry	10
6.3 Error_check	11
6.4 Ancillary_data	11
7 Method of signaling the audio coder control data bit stream on the AES3 interface	11
Annex A (informative) AES41 metadata types	13
Annex B Bibliography	14

Foreword

[This foreword is not a part of *AES standard for digital audio — Recoding data set for audio bit-rate reduction*, AES41-2000.]

This document describes a detailed specification of a set of coding-decision data for MPEG Layer II with a generic transport mechanism developed as project AES-X78 in the AESSC SC-02-02 Working Group on Digital Input-Output Interfacing of the SC-02 Subcommittee on Digital Audio. The task group was lead by A. Mason. The intent has been to produce a standard in cooperation with other bodies. Individual experts associated with potential end-users, equipment developers, and manufacturers who are directly and materially affected by this project were encouraged to participate in the process of standardization by joining SC-02-02.

In the near future, digital compression techniques are expected to dominate the broadcast television environment. Digital compression will be used for a significant amount of program acquisition, for program production and storage, and for distribution and broadcasting.

Compression systems add impairments to sound. The audibility of these impairments depends on the degree of compression and the techniques used. It has been proved that re-encoding with minimal further impairment can be achieved by using information describing the previous encoding process. The information that is preserved varies according to the encoding scheme that has been used. For example, for transparent cascaded coding of MPEG-1 Layer II or MPEG-2 Layer II audio, the information required by the downstream encoder includes:

- . the positions of the frame boundaries;
- . the bit rate of the compressed bit stream before decoding;
- . the coding mode (monophonic, joint stereo, and so on);
- . the bit allocation for each sub-band within a frame;
- . the scale factors for each sub-band within each sub-block of a frame.

In order to help with the process of editing decoded audio, it is useful to send additional coding-decision data to describe any timing offset that may be introduced by editing only at points corresponding to block boundaries (for example, 24-ms boundaries for MPEG-1 Layer II at 48 kHz). These data are particularly relevant for editing sound associated with co-timed video. The bit rate of these additional data is less than 30 kbit/s. A cyclic-redundancy check word must be appended to frames of data to provide a means to detect errors in the data.

This data signal can be used by a downstream encoder to guide its coding decisions. For example, if the audio signal is changed in any way (as it would by mixing it with another signal) then it could be inappropriate for the downstream encoder to re-use any previous coding decisions. However, in such a case, when a suitable transport mechanism is used, the coding-decision data would be corrupted so the coder could detect that the data are no longer valid.

This specification describes in detail the content of the coding-decision information, such as MPEG-header information, bit allocations, and scale factors, plus its representation in terms of the number, order, and meaning of its bits. A mechanism by which these data can be conveyed in a pulse-code-modulation audio signal is also described. The mechanism modifies one of the least significant bits of the audio word to signal the data. The precise time relationship of the data to the audio is also defined.

R. Finger, chair, SC-02-02
J. Dunn, vice-chair, SC-02-02
2000-04-08

Foreword to the second edition, 2009

As predicted in the foreword to this document in 2000, digital compression techniques now dominate the broadcast television environment. In addition to the problems foreseen relating to cascaded compression, new problems have arisen because of the use of loudness control and surround sound with those digital compression techniques.

Metadata within the compressed audio bit-stream is used to control loudness and the mixing down of multi-channel surround sound to two-channel stereo. These metadata are usually known by terms such as "dialnorm", "prog_ref_level", and "down-mix coefficients".

Whilst this might seem unrelated to the original scope of AES41, dealing with bit allocations and scale factors, it is simply another form of data that can affect a later encoding of the audio: this time it is more macroscopic than microscopic.

The metadata is lost when the bit-stream is uncompressed unless provision is made to transport it or store it somewhere. Existing methods rely on non-audio mechanisms to convey the metadata alongside the audio, for example a serial data link like RS-422 and serial digital video SMPTE 259M, or a "chunk" in an audio file (for metadata that does not change).

This revision extends AES41 to include data formats for carrying this loudness and down-mix metadata with the uncompressed PCM using the same transport mechanism as before. The metadata can therefore be carried in the audio to which it relates.

J. Grant
Chair, SC-02-02 Working Group on digital input/output interfacing
2009-12

Foreword to the third edition, 2012

The proposal to add a further data type to this standard led the working group to conclude that this should be done in conjunction with a partitioning of the standard. The resulting multi-part structure provides a clear separation between the specification of the mechanism by which data is carried and the specification of the semantics of the data that is carried. This simplifies the use of the standard since a reader might not need all the parts, and it simplifies the addition of further data types in the future.

Please note that clause numbers in this edition of the standard will, in general, no longer correspond to clause numbers in the first or second editions.

The draft of this document was developed by a writing group whose primary author was Andrew Mason.

J. Grant
Chair, SC-02-02 Working Group on digital input/output interfacing
2012-03

Note on normative language

In AES standards documents, sentences containing the word "shall" are requirements for compliance with the document. Sentences containing the verb "should" are strong suggestions (recommendations). Sentences giving permission use the verb "may". Sentences expressing a possibility use the verb "can".

AES standard for digital audio - Audio-embedded metadata - Part 1: General

0 Introduction

0.1 Rationale for this standard

It has become apparent that there are requirements for the transport of audio metadata together with the related audio, in a way such that the two are not separated. Further, there are requirements where any alteration of the audio should render the related metadata invalid. Metadata held in files, or transported in associated video signals, or in AES3 user bits, for example, does not meet these requirements: The metadata may become dissociated from the audio, or may continue to be associated with the audio even after substantial changes have been made.

Initially, this standard was developed to satisfy these requirements, of tight association and fragility, for an application in cascaded digital compression (bit rate reduction). Other applications have been identified that have these same two requirements, including the carriage of metadata to describe loudness.

0.2 Conventions used in this standard

0.2.1 Decimal points

According to IEC directives, the comma is used in all text to indicate the decimal point. However, in the specified coding, including the examples shown, the full stop is used as in IEC programming language standards.

0.2.2 Data representation

In this standard, all coding and data representations are printed in an equally spaced font.

0.2.3 Non-printing ASCII characters

Non-printing characters are delimited by angle brackets, as in <CR> for carriage return.

0.3 Patents

The Audio Engineering Society draws attention to the fact that it is claimed that compliance with this AES standard may involve the use of an application for patents concerning "Lip sync" and "Lip-sync with sub-frame error feedback."

The AES holds no position concerning the evidence, validity and scope of this patent right.

The holder of this patent right has assured the AES that it is willing to negotiate licenses under reasonable and non-discriminatory terms and conditions with applicants throughout the world. In this respect, the statement of the holder of this patent right is archived with the AES.

Information may be obtained from:

British Broadcasting Corporation
Attention of Daniel Pike
BBC Research and Development
Centre House
56 Wood Lane
London, W12 7SB
UK

Attention is drawn to the possibility that some of the elements of this AES standard may be the subject of patent rights other than those identified herein. AES shall not be held responsible for identifying any or all such patent rights.

1 Scope

This document describes a mechanism for embedding a set of data in an audio signal. The syntax and semantics of a number of sets of data are defined in other parts of this standard. Provision is also made for the transmission of additional ancillary information. The data is embedded in the audio so that it is transported with the audio so that the data may be recovered and used to control subsequent processing of the audio, for example by low bit-rate coders or dynamics processors.

2 Normative references

The following standards contain provisions that, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the indicated standards.

This standard has no normative references

3 Definitions, symbols, and abbreviations

3.1 Definitions

3.1.1

CRC

cyclic redundancy check

3.1.2

nextbits function

`nextbits()`

function that permits comparison of a bit string with the next bits to be decoded in the bit stream

3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

3.2.1 Arithmetic operators

3.2.1.1

`+`

addition

3.2.1.2

`++`

increment

3.2.1.3

`--`

decrement

3.2.1.4

*

multiplication

3.2.1.5

%

modulus operator, defined only for positive numbers

3.2.2 Relational operators

3.2.2.1

<

less than

3.2.2.2

==

equal to

3.2.2.3

!=

not equal to

3.2.3 Bitwise operators

&

AND

3.2.4 Assignment

=

assignment operator

3.3 Abbreviations

The following mnemonics are defined to describe the different data types used in the coded bit stream.

3.3.1

bslbf

bit string, left bit first, where "left" is the order in which bit strings are written in this standard.

NOTE Bit strings are written as a string of logic 1's and logic 0's within single quote marks, for example, '1000 0001'. Blanks within a bit string are for ease of reading and have no significance.

3.3.2

rpchof

remainder polynomial coefficients, highest order first

3.3.3

simsbf

signed integer, most significant bit first

3.3.4

uimsbf

unsigned integer, most significant bit first

4 Method for describing the bit-stream syntax

The bit stream containing the audio coder control data shall be as described in 5. Each data item in the stream shall be described by its name, its length in bits, and a mnemonic for its type and order of transmission.

NOTE 1 In the following, constructs used to express the conditions when data elements are present are in non-bold type.

NOTE 2 This syntax uses the C-language convention that a variable or expression evaluating to a non-zero value is equivalent to a condition that is true.

<pre>If(condition){ data_element ... } else { data_element ... }</pre>	If the condition is true, then the first group of data elements occurs next in the data stream. If the condition is not true, then the second group of data elements occurs next in the data stream.
<pre>For(expr1;expr2; expr3){ data_element ... }</pre>	Expr1 is an expression specifying the initialization of the loop. Normally it specifies the initial state of the counter. Expr2 is a condition specifying a test made before each iteration of the loop. The loop terminates when the condition is not true. Expr3 is an expression that is performed at the end of each iteration of the loop. Normally it increments a counter.

The most common usage of this construct may be as follows:

<pre>for(i=0; i<n; i++) { data_element ... }</pre>	The group of data elements occurs n times. Conditional constructs within the group of data elements may depend on the value of the loop control variable i, which is set to zero for the first occurrence, incremented to one for the second, and so forth.
---	--

As noted, the group of data elements may contain nested conditional constructs. For compactness, the `{ }` may be omitted when only one data element follows.

<code>data_element[]</code>	<code>data_element[]</code> is an array of data. The number of data elements is indicated by the context.
<code>data_element[n]</code>	<code>data_element[n]</code> is the $(n+1)$ th element of an array of data.
<code>data_element[m][n]</code>	<code>data_element[m][n]</code> is the $(m+1,n+1)$ th element of a two-dimensional array of data.

5 Specification of the audio coder control data bit-stream syntax

The semantics of the elements in the following tables is explained in 6.

Syntax	No. of bits	Mnemonic
<pre>data_sequence() { while(nextbits()==synchronization_word){ data_frame() } }</pre>		

Syntax	No. of bits	Mnemonic
<pre>data_frame() { header() data() error_check() ancillary_data() }</pre>		

Syntax	No. of bits	Mnemonic
<pre>header() { synchronization_word length reserved, 00 introduced_time_offset_flag type introduced_time_offset }</pre>	32 8 2 1 5 16	bslbf uimsbf uimsbf bslbf uimsbf simsbf

Syntax	No. of bits	Mnemonic
<pre>data() { the format of each type of data, as indicated in the header, is defined in other Parts of this Standard }</pre>		

Syntax	No. of bits	Mnemonic
<pre>error_check() { crc_check }</pre>	16	rpchof

Syntax	No. of bits	Mnemonic
<pre> ancillary_data() { for(b=0; b<no_of_ancillary_bits;b++) ancillary_bit } </pre>	1	bslbf

6 Semantics of the bit-stream syntax

The **data_frame** syntax shall be as described in this clause.

header	Part of the bit stream containing synchronization and content indicators.
data	Part of the bit stream containing information about the audio signal (for example, previous encoding, loudness information).
error_check	Part of the bit stream containing information for error detection.
ancillary_data	Part of the bit stream that may be used for ancillary data.

6.1 Header

The first 54 bits constitute the header information which is common to all audio coder control data types, where:

synchronization_word	is the bit string '0000 0000 0000 0000 0000 0001 1111 1011';
length	is the number of bytes (octets) in the data_frame following the length count, up to and including the last bit of the CRC, whereby padding_bits in the recoding_data ensure that the length is a multiple of 8 bits;
type	indicates the type of data that follow. See 6.2 and annex A;
flags	are bits indicating the validity of fields in the data frame header;
introduced_time_offset_flag	is a 1-bit flag, which when set to logic 1 indicates that the introduced time offset field in the header is valid; when set to logic 0 it indicates that the introduced time offset field is not valid (in which case the introduced_time_offset should be set to logic 0);
introduced_time_offset	indicates a time offset, measured in sample periods, by which the audio signal has been shifted with respect to a reference time. A positive offset represents a delay, a negative offset indicates that the audio signal is in advance of the reference time. The time offset is represented by a signed integer with the most significant bit inverted to reduce the probability of imitation of the synchronization word.

6.2 AES41 metadata type registry

AES41 metadata types are assigned and registered by the AES. A current list of assigned types is published on the AES standards web site. Unassigned values are reserved for future AES standardization. See also annex A.

The type logic 0 (undefined) should be used in situations where there is no knowledge of a previous encoding scheme, including those where the audio has not yet been encoded, or this transport mechanism is being used to carry data that is not one of the standardised types.

If an audio channel can only transport the sixteen most significant bits of the AES3 sample value, then the data should be carried by having the sixteenth most significant bit set so that the parity of the sixteen most significant bits is as required.

If the audio channel can transport twenty or more bits transparently, then the data should be signaled by having the twentieth most significant bit set so that the parity of the twenty most significant bits is as required.

If the audio channel can transport 24 or more bits transparently, then the data should be signaled by having the twenty-fourth most significant bit set so that the parity of the twenty-four most significant bits is as required.

If a monophonic audio signal is being carried on one channel of an AES3 interface, the data should be signaled in the same channel as the audio signal is being carried.

If an audio signal consisting of two channels (either stereophonic or independent) is being transmitted on an AES3 interface, separate data should be signaled for the two channels. The data for each channel should be sent in the channel in which the signal to which it relates is being transmitted.

Channel status data shall indicate that the interface is being used for audio.

Annex A (informative) AES41 metadata types

This standard provides for assignment and registration of AES41 metadata types - see 6.2. A current list of assigned types is available at: www.aes.org/standards/data/registered-values.cfm

Registered types current at the time of publication of this Part 1 are:

Table A.1 - Metadata types

Type	Encoding scheme
0 ₁₀	Undefined (see 6.1)
1 ₁₀	MPEG-1 Layer I or MPEG-2 low sampling frequency (LSF) Layer I
2 ₁₀	MPEG-1 Layer II or MPEG-2 LSF Layer II
3 ₁₀	MPEG-1 Layer III or MPEG-2 LSF Layer III
4 ₁₀	ISO/IEC 14496-3:2005 minimal metadata
5 ₁₀	Dolby E audio minimal metadata
6 ₁₀	EBU loudness, true-peak, and downmix metadata
7 ₁₀ to 31 ₁₀	Reserved for future standardization

Annex B Bibliography

ISO/IEC 11172-3 (1993-08), *Information technology - Coding of moving pictures and associated audio for digital storage media at up to about 1,5 Mbit/s - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

ISO/IEC 13818-3 (1998-05), *Information technology - Generic coding of moving picture and associated audio information - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

ISO/IEC 14496-3:2005(E), *Information technology – Coding of audio-visual objects - Part 3: Audio*. International Electrotechnical Commission, Geneva, Switzerland.

ITU-R Recommendation BS.1770-1, *Algorithms to measure audio programme loudness and true-peak audio level*, International Telecommunications Union, Geneva, Switzerland.

SMPTE RDD 6-2008 *Registered Disclosure Document: Television – Description and Guide to the Use of the Dolby E Audio Metadata Serial Bitstream*. Society of Motion Picture and Television Engineers, White Plains, NY., US.